Malicious URL Detection with Explainable Machine Learning Techniques

Bolun Wang*
School of Cyberspace
Hangzhou Dianzi University
Hangzhou, Zhejiang, China
1908333700@qq.com

Abstract

Malicious Uniform Resource Locator (URL) detection research has been actively worked on and the focus is on achieving higher and more accurate results with machine learning and artificial intelligence approaches. Unfortunately, existing research on this problem has not yet been completed completely, and there remains a research gap in malicious URL detection in developing effective approaches that are capable of dealing with evasive techniques used by the attackers to conceal harmful URLs. This study explores an effective technique of detecting malicious URL detection with machine learnings with explainability. In particular, three advanced ML models are applied on one real parameters URL dataset, Logistic regression (LR), decision trees (DT) and Random Forest (RF) are employed. The results show that RF has the best detection accuracy and the best performance in terms of explain ability.

CCS Concepts

• **Information systems** → World Wide Web; Web interfaces.

Keywords

Malicious URL Detection, Machine Learning, Explainability

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1 INTRODUCTION

With advancing internet technology, plethora of businesses that are online, for instance, social networking, digital payments, and e commerce, are experiencing incredible growth [1]. There is a shift in major enterprises and public institutions to online platform of operation where their services are offered, which has led to increasing efficiency and transparency of operations [2].

Numerous convenience brought by technological advancements, but they also prepare sophisticated cyber attack and the deceptive techniques against users [3]. Malicious activities, appear in as many different forms as selling merchandise from phony websites, financial fraud schemes, and malware distribution. This is accomplished through a wide variety of methods such as explicit hacking, SQL injection, phishing, watering hole attacks, man-in-the-middle attacks, device loss or theft, denial of service and distributed denial of service attacks. With a number of attack vectors and wide complexity of the cybersecurity landscape, it is a challenging problem to design effective systems that can detect and defend against these threats [4].

Most malicious URLs are applied as a common cyber attack, and statistics show almost one third of web sites are intended for mal [5]. The sites send out unwanted content via spam, phishing and through drive by downloads. However, visiting such sites lures users as victims to various risks. Drive by downloads, phishing, social engineering, spam are among the common attack methods, all of them presenting serious and unique threats. By advertising spams or phishing attempts, spam emails may result in major losses from the victim user [6]. It is essential to detecting malicious URLs reduces network security risks. In order to counter different network attacks, a good detection system can be implemented and provide users big security protection of their property and information. But as researchers and cybersecurity professionals keep developing more efficient solutions to get the detection accuracy and defense capabilities up, the user shall always be safe [7].

Older methods are based on blacklist filtering and rule based matching. The blacklist filtering approach consists of maintaining a database with blacklisted URL's that were already identified as malicious ones. During detection, this database is checked to see if the URL is present. However, the rule based matching approach will detect the malicious URLs using predefined rules and patterns, for example like regular expressions, string patterns or behavioral features. This is manually crafted by security experts, and updated on regular basis.

Although both methods can work in some situations, they are greatly limited. With respect to the URLs not yet in the database, blacklist filtering is ineffective due to the need of frequent database updates to counter of newly emerging malicious URLs. Defining and maintaining a large set of rules is complex and error prone when doing rule based matching. False Positives and missed detections are a result of improper configurations of rules. Therefore, it is challenging to design and maintain an efficient rule base, which demands substantial resources.

*Corresponding author.



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Machine learning has demonstrated significant success in engineering applications by enabling automated control of complex tasks through advanced analysis of large datasets, extracting actionable patterns to optimize operations, detect anomalies, and enhance efficiency across diverse domains [8].

This paper evaluates and compared three machine learning classifiers such as Logistic regression (LR), decision trees (DT), and Random Forest (RF) based on a real-world malicious URL detection dataset where RF is shown to be most effective detection. The explainability of the RF classifier is evaluated in this paper as well, to improve the usability of machine learning classifier in a real world detection system.

2 RELATED WORK

2.1 Traditional Methods

Existing blacklisting and rule matching practises to identify malicious URLs are in widespread practise however arrive with their own limitations. While blacklisting filtering blocks the new malicious URLs, it requires continuous updates to keep effectiveness over the newly coming malicious URLs. In rule based matching, we have to define and maintain a complex set of rules which can lead to errors in implementation as well as may result in false positives or do not detect at all if not properly configured. However, these challenges in traditional approaches have not been remedied without intervention according to machine learning techniques which complement and supplement traditional methods to improve the accuracy of detection and make cybersecurity go adaptive towards dynamic threats.

Identifying key features that vary legitimate URLs from phishing URLs is explored by Jeeva et al. [9]. A priori and predictive association rule mining are then used to analyze these features. The resulting rules reveal commonalities that are present in phishing URLs. The features, such as transport layer security, lack of top level domains in the URL, and specific keywords in the URL path are sensitive features that indicate their phishing URLs.

The limitations of sum-lists methods in detecting new web sites are discussed by Rafsanjani et al. [10]. Although machine learning methods have gained popularity for URL malicious detection improvement, they depend greatly on the data and need a large volume of datasets which should be normally updated to be effective for training. QsecR, an Android Secure QR code Scanner, which places priority on privacy, is introduced in this study. A predefined static feature classification with 39 categories is used by QsecR.

This research reveals several drawbacks with regard to the existing methods in detecting malicious URLs. Blacklist filtering and rule based matching are methods that require frequent update in order to be able to respond to the newly emerging threat. Both blacklist filtering may not work against URLs which are not yet listed in its database and rule based matching requires an error prone and complex definition and maintenance of detection rules. In the event of improperly configured rules, ensures that accuracy and adaptability to ever changing cyber threats would be maintained accurately.

2.2 Machine Learning Methods

URL detection is formulated into a binary classification task and machine learning methods are used to train models by using extracted features from URLs. Research indicates that machine learning approaches can significantly enhance detection accuracy by autonomously identifying unknown malicious URLs based on learned patterns. However, these methods rely heavily on the data quality, requiring large, up-to-date datasets to effectively generalize across various types of cyber threats. Ongoing research aims to refine machine learning models for better performance in real-world scenarios and to mitigate challenges such as adversarial attacks and data privacy concerns.

Rashid [11] points out that most phishing attempts are short-lived and frequently use newly created domains. As a result, existing machine learning and deep learning methods for detecting phishing URLs are only effective if the test set comes from the same dataset as the training set. To overcome this limitation, the study proposes a framework based on unsupervised domain adaptation, which aims to improve model portability and effectiveness across different datasets.

In the study by Karim et al. [12], a machine learning approach is used to address the increasing threat of phishing attacks. The researchers apply various methods, including hybrid LSD models, decision trees, and linear regression, to a phishing URL dataset, demonstrating their effectiveness in combating phishing. Notably, the LSD model-integrating LR, SVM and DT-proves to be the most effective among the models tested.

In the study by Abdul Samad et al. [13], machine learning models are used to identify phishing URLs and analyze the complexity of phishing attacks. In this way these models evaluate the URL, web content and site functionality from external point of view. Various tuning factors figured to make accuracy less (e.g., data balancing, hyperparameter optimization, feature selection) are found to lead to significant accuracy. Through this combination, these factors made more advanced results of machine learning techniques. The accuracy rates for Random Forest method are 97.44% and Gradient Boosting 97.47% for Dataset-1.

Biswas et al. [14] develop a hybrid framework that leverages on explainable AI for analyzing cyber risks on interconnected phishing attacks. The framework includes several steps: identify the skills of phishers, estimate the phishing attack likelihood in presence of existing IT security, use machine learning classifiers to classify URLs, infer the joint distribution using exponential beta models, and utilize Archimedes copula to calculate the potential losses from phishing attacks. The results of this approach provide a company with insights as to their risk tolerance, so that cybersecurity investments or cyber insurance can be better allocated with this gained knowledge.

Khan et al. [15] compare the effectiveness of various machine learning methods-DT, SVM, RF, Naive Bayes (NB), kNN, and Artificial Neural Networks (ANN)-across three different datasets. They also evaluate these models on a dimensionality reduced version of the dataset. Random Forest and Artificial Neural Network achieve an accuracy of 97% as per results from UCI machine learning repository as well as other sources.

Table 1: Main domain features.

Feature Number	Feature	Explanation
1	pdomain_len	Length of the primary domain name
2	pdomain_count_dot	Number of "." in the primary domain name
3	pdomain_count_hyphen	Number of "-" in the primary domain name
4	pdomain_count_atrate	Number of "@" in the primary domain name
5	pdomain_count_non_alphanum	Number of non-alphabetic and non-numeric characters in the primary domain name
6	pdomain_count_digit	Number of digits in the primary domain name
7	pdomain_min_distance	Minimum distance to the primary domain, used to measure similarity to known domains

According to Salihovic et al. [16], the UCI phishing and spam datasets are used to analyse the effect of feature selection on different machine learning algorithms. They test six algorithms: RF, kNN, ANN, SVM, LR, and NB. Results show that different feature selection methods yield varying performance levels. Specifically, Ranker combined with Principal Component Optimization achieves 97.33% accuracy with Random Forest, while BestFirst combined with CfsSubsEval Optimization results in 94.24% accuracy for the same model.

Vishva et al. [17] propose a phishing detection system that uses both URL features and web page content features. The UCI phishing dataset is used to extract URL features, and the Term Frequency-Inverse Document Frequency (TF-IDF) method generates features from web content. Three machine learning algorithms, LR, RF and SVM have evaluated as URL classification. The results show that Random Forest algorithm has the highest classification accuracy with 96%.

There are several research challenges in using machine learning methods for detecting malicious URLs. First of all, these methods also heavily depend on large and many diverse datasets that must be trained on for perfecting the model, which can be hard to get and may not always cover all the threats. On top of that, such machine learning models are vulnerable to adversarial attacks in which an adversary can poison feature values to trick the model in avoiding detection. A huge obstacle lies ahead in guaranteeing that these models generalize strongly across various types of malicious URLs, as these models might confuse to the patterns specific to the same in the training data, thus exhibiting poor performance on the unobserved threats. Solving these issues is crucial to enable the deployment of machine learning based solutions in cyber security practice as effective and as reliable.

3 Methodology

3.1 Dataset Description

This study used the Kaggle dataset (https://www.kaggle.com/datasets/pilarpieiro/tabular-dataset-ready-for-malicious-url-detection) for malicious URLs detection and to further research in the field of cybersecurity. A complete set of both benign and harmful URLs together with a great variety of features. Basic components, domain details, content analysis, path and query parameters, SSL certificate information, host reputation, network characteristics, machine learning derived metrics, and behavioral insights are some of these features which involve URL based aspects.

The rich set of data presented helps researchers and data scientists accrue and evaluate URL detection models. The dataset is very useful to increase the accuracy and effectiveness of URL classification systems and enhances the collaboration and knowledge sharing in the area of cybersecurity.

Table 1 displays the main domain features required to describe the dataset, Table 2 presents the overall URL features and Table 3 presents the top-level domain features.

3.2 Classifier Description

Three machine learning classifies were used in this study, namely, DT, RF and LR.

Decision tree has branches that lead to a decision node or a leaf node, which is an output or class label. The decision rule for each branch is represented by each branch, and the hierarchical structure of the tree facilitates easy interpretation and understanding of the effect of different factors on the final prediction.

Random forest trains many decision trees on the data subset and aggregates their predictions by voting or simple averaging. Random forest borrows the variety of trees to improve the performance. It supplies a good answer to classification and regression issues, as it cuts down on variance and increasing stability than on the individual decision tree, meanwhile, it is less inclined to overfitting.

The logistic regression fits the predicted values using a logistic function (or sigmoid function) mapping to a range of probability 0,1. This method computes the probability based on a set of inputs and one particular outcome and is appropriate for use when the dependent variable is categorical.

4 Results and Discussion

4.1 Settings

Then, it splits the data into training set and test set with ratio of 75: 25. Specifically, this means that 75 percent (the entire available dataset) would be shipped to train, and 25 percent of the same dataset would be sent for testing. The training set contains 6,728,848 samples, which include 5,283,175 benign ones. The test set is composed of 1,682,213 samples with 1,320,794 samples being benign. Python is used to implement the machine learning models and accuracy, precision, recall and F1 score are the evaluation metrics.

Table 2: Overall URL features.

Feature Number	Feature	Explanation
1	url_has_login	Whether to include the keyword "login" or not, 1 means include, 0 means don't include.
2	url_has_client	Whether to include the keyword " client " or not, 1 means include, 0 means don't include.
3	url_has_server	Whether to include the keyword " server " or not, 1 means include, 0 means don't include.
4	url_has_admin	Whether to include the keyword " admin" or not, 1 means include, 0 means don't include.
5	url_has_ip	Whether to include the IP address, 1 means include, 0 means not include
6	url_isshorted	Whether it is a short URL, 1 means yes, 0 means no
7	url_len	The length of the URL, the total number of characters in the URL
8	url_count_dot	Number of "." in URL
9	url_count_www	Number of "www" in URL
10	url_count_atrate	Number of "@" in URL
11	url_count_exclamation	Number of "ঙ" in URL
12	url_count_space	Number of spaces in the URL
13	url_count_tilde	Number of "~" in URL
14	url_count_comma	Number of "," in URL
15	url_count_plus	Number of "+" in URL
16	url_count_asterisk	Number of "*" in URL
17	url_count_hashtag	Number of "#" in URL
18	url_count_dollar	Number of "$" in URL
19	url_count_percent	Number of "%" in URL
20	url_count_and	Number of "&" in URL
21	url_count_hyphen	Number of "-" in URL
22	url_count_underscore	Number of "_" in URL
23	url_count_equal	Number of "=" in URL
24	url_count_question	Number of "?" in URL
25	url_count_digit	Number of digits in the URL
26	url_count_alpha	Number of letters in the URL
27	url_count_non_alpha	Number of non-alphabetic characters in the URL
28	url_count_non_alphanum	Number of non-alphabetic and non-numeric characters in URL

Table 3: Top-level domain features.

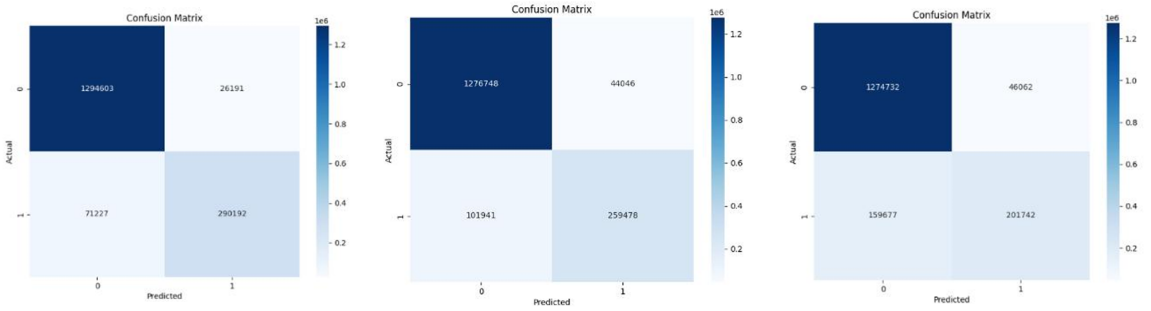
Feature Number	Feature	Explanation
1	tld_len	Length of Top Level Domain (TLD)
2	tld	Top-level domains, e.g. de, com, com.cn, etc.
3	tld_is_sus	Whether the TLD is suspicious, 1 means suspicious, 0 means not suspicious

Table 4: Comparison of different classifiers.

Classifier	Accuracy	Precision	Recall	F1 Score
DT	0.913	0.855	0.718	0.780
RF	0.942	0.917	0.803	0.856
LR	0.877	0.816	0.556	0.661

4.2 Results

Table 4 shows the comparison of different classifiers, in terms of accuracy, precision, recall and F1 score. The best performance based on the F1 score is achieved by the RF classifier and the results are better than the rest of the models.



(a) RF (b) DT (c) LR

Figure 1: The confusion matrices.

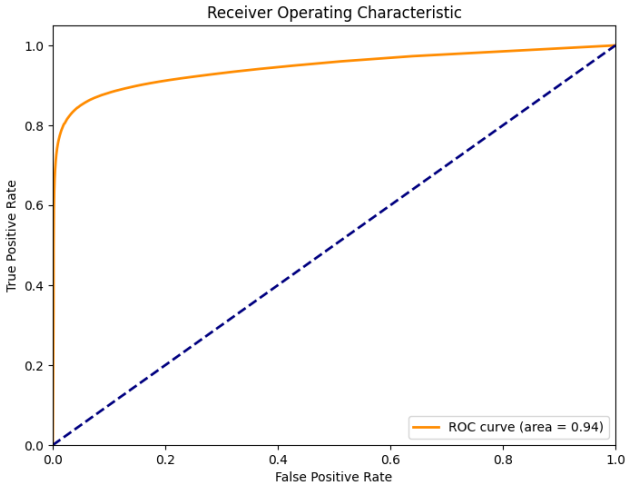


Figure 2: The ROC curve of the RF classifier.

The confusion matrices are further shown in Figure 1. The classification results are presented in terms of a receiver operation characteristic (ROC) curve of the RF classifier in Figure 2, demonstrating an effective detection performance in malicious URL classification.

Feature importance scores of the RF classifier are calculated and plotted to evaluate the explainability of the RF classifier in Figure 3. In the above experiment we prove that url_len, url_entropy and path_len are the most important three features in detection of malicious URL.

The RF classifier is finally improved to increase its explainability through an implementation of the SHAP (SHapley Additive exPlanations) tool. SHAP is an interpretation tool based on Shapley value theory to explain the how a machine learning model outputs how to assign a feature importance value to each given feature in the model. The Shapely value is borrowed from the SHAP value used to calculate and fairly to apportion the value that the cooperation adds to multiple player. In machine learning model interpretation, this concept is used in SHAP, calculated feature contribution to the model's predicted outcome with a local and global interpretation of model. The definitions of consistency, local interpretation and global interpretation hold true for the characteristics of SHAP. This implies that if model feature contribution increases SHAP value will also increase by the same proportion. Describe local interpretation of some predictions and global interpretation of the model as a whole, through aggregation of SHAP values over many points of data. I In order to calculate the effect of each feature on each data point, the use of SHAP values is in form. SHAP is sufficiently widely used to rank feature importance, interpret individual predictions, and also discover anomalies.

In Figure 4 we have the SHAP values of several features of the RF classifier. As shown in Figure 3, url_len, url_entropy and path_len are still very contributing in classes. Nevertheless, from Figure 4, path_count_no_of_dir and subdomain_len are the two most important features.

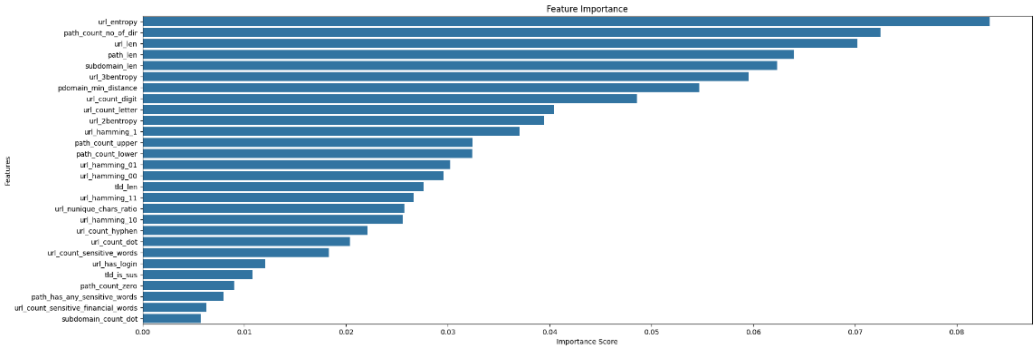


Figure 3: The feature importance scores of the RF classifier.

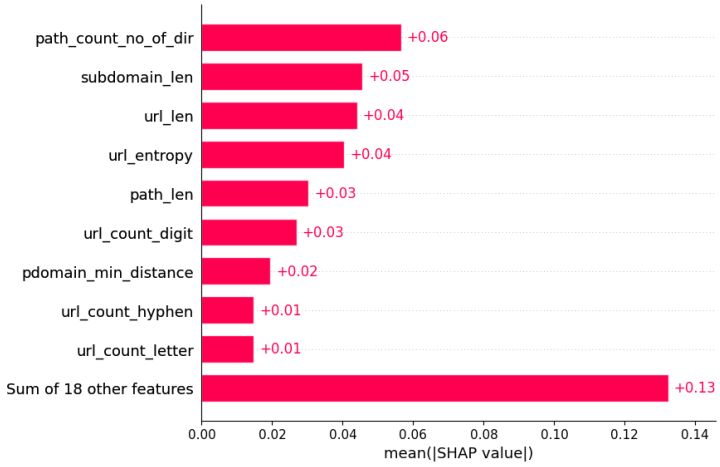


Figure 4: SHAP values for different features of the RF classifier.

5 Conclusion

This paper provides a comprehensive evaluation and comparison of three machine learning classifiers namely, LR, DT and RF that are based on a real-world malicious URL detection dataset, where RF has been proven to be the most suitable detection solution. Furthermore, this paper evaluates the explainability of the RF classifier in order to show the usefulness of the method.

Research on malicious URL detection in the future is possibly to be used on model robustness and adaptation to evolving threats. Some of the areas where research can be carried out are integrating the most advanced machine learning techniques to significantly enhance detection accuracy and leverages for complex attack vectors. In addition, multi modal data sources such as combining URL features with behavior analytics, network traffic patterns etc. are also being investigated by the researchers for more sophisticated threats detection. The second important direction is the development of the real-time detection systems, which can keep pace with changes in froms of threats. Further, it is key to increase the interpretability and explainability of the detection models in order to aid in better decisions and trust in automated systems. Malicious URL detection technology will need to rely on collaboration across the cybersecurity community to share insights and broaden the whole dataset.

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